

razor__data__notebook

July 6, 2026

1 6/26 Razor Test 1

1.0.1 Notes:

Collimator face to reflection off wedge 95mm

Wedge to mirror 135mm

Mirror to lens 142mm

Lens to blade 66 mm

Blade to sensor 200mm

Focal length 50mm 1in dia

Computational dist: | d1: 326.57 (320.5 face to d1) | d2: 55.32 | Path: 375.82

Measured distances: | d1: (372) | d2: 66 | Path: 438

```
[1]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from scipy.optimize import curve_fit
from scipy import signal
from scipy.signal import savgol_filter, medfilt
```

1.1 Smoothing Algorithms

1. **Moving Average (MA)** — (Codemonk, *Moving Average Filter*)
2. **Exponential Smoothing (SES/EWMA)** — (InfluxData, *Exponential Smoothing Guide*)
3. **Savitzky-Golay (SG)** — (Konstantinovskiy, *Savitzky-Golay Filter Guide*)
4. **Median Filter** — (median_filtering reference)
5. **FFT-based denoising** — (Lin et al. 2023, *NIM A* — FFT-based beam profile denoising)

```
[2]: # Moving Average Filter is a Finite Impulse Response (FIR) Filter used for
    ↪ smoothing
    # the signal and reducing noise.

    # Due to having a fixed window this may be used on both the S-curve and gaussian
    ↪ however
    # edge cases get screwed up.
    def simple_moving_average(signal, window=15):
        return np.convolve(signal, np.ones(window)/window, mode='same')

    # Making sure to handle edge cases
```

```
def moving_average_filter(signal, window=15):
    kernel = np.ones(window) / window
    pad = window // 2
    padded = np.pad(signal, pad, mode='edge')
    smoothed = np.convolve(padded, kernel, mode='same')
    return smoothed[pad:pad + len(signal)]
```

```
[3]: # Exponential Smoothing / EWMA
#  $s(t) = \alpha * x(t) + (1 - \alpha) * s(t-1)$ 

def exponential_smooth(signal, alpha=None):
    model = SimpleExpSmoothing(signal, initialization_method="estimated")
    fit = model.fit(smoothing_level=alpha, optimized=(alpha is None))
    return fit.fittedvalues, fit.params['smoothing_level']
```

```
[4]: # Savitzky-Golay Filter
# Fits a polynomial to each window by least squares.
# Window needs to be an odd number.

def sg_smooth(signal, window_length=25, polyorder=3):
    if window_length % 2 == 0:
        window_length += 1
    return savgol_filter(signal, window_length=window_length,
        ↪polyorder=polyorder)
```

```
[5]: # Median Filter
# Each point is replaced by the median of an odd-sized neighborhood window.

def median_smooth(signal, window=9):
    if window % 2 == 0:
        window += 1
    return medfilt(signal, kernel_size=window)
```

```
[6]: # FFT-based denoising (Lin, Tan, Li, Wang & Shen, 2023, NIM A 1047:167781)
# I. FFT the (uniformly resampled) signal, shift zero-frequency to center.
# II. Estimate the low-frequency Gaussian width  $\sigma_{lf}$  from the magnitude
# spectrum's FWHM, and discard all frequency content beyond  $4*\sigma_{lf}$ .
# III. Within the retained band, drop any remaining low-amplitude components
# below a noise threshold  $3*\sigma_{hf}*\sqrt{2*\log_{10}(N)}$ , where  $\sigma_{hf}$  is
# std. of the amplitudes in the discarded high-frequency band.
# IV. Inverse FFT back to the spatial domain.
# This assumes evenly-spaced samples and a Gaussian signal.

def sigma_lf_fwhm(freqs, magnitude):

    center_index = len(freqs) // 2
    peak = magnitude[center_index]
    half_max = peak / 2
    right_index = center_index
```

```

while right_index < len(freqs) - 1 and magnitude[right_index] > half_max:
    right_index += 1
if right_index > center_index:
    f0, f1 = freqs[right_index - 1], freqs[right_index]
    m0, m1 = magnitude[right_index - 1], magnitude[right_index]
    f_half = f0 + (half_max - m0) * (f1 - f0) / (m1 - m0) if m1 != m0 else f1
else:
    f_half = freqs[right_index]
fwhm = 2 * abs(f_half - freqs[center_index])
return fwhm / 2.355 # FWHM -> sigma for a Gaussian

def fft_denoise_profile(x_raw, profile_raw):
    n = len(x_raw)
    x_uniform = np.linspace(x_raw.min(), x_raw.max(), n)
    # FFT needs even spacing and linear interpolation should be fine following
    → that it
    # should register as high frequency noise and get filtered out.
    p_uniform = np.interp(x_uniform, x_raw, profile_raw)
    dx = x_uniform[1] - x_uniform[0]

    spectrum = np.fft.fftshift(np.fft.fft(p_uniform))
    freqs = np.fft.fftshift(np.fft.fftfreq(n, d=dx))
    magnitude = np.abs(spectrum)

    sigma_lf = sigma_lf_fwhm(freqs, magnitude)
    → # Step I / II
    lf_mask = np.abs(freqs) <= 4 * sigma_lf
    spectrum_filtered = spectrum * lf_mask

    hf_magnatudes = magnitude[~lf_mask]
    → # Step III
    sigma_hf = np.std(hf_magnatudes)
    N = len(hf_magnatudes)
    threshold = 3 * sigma_hf * np.sqrt(2 * np.log10(N))
    spectrum_filtered[np.abs(spectrum_filtered) < threshold] = 0

    denoised_uniform = np.real(np.fft.ifft(np.fft.ifftshift(spectrum_filtered)))
    → # Step IV
    denoised = np.interp(x_raw, x_uniform, denoised_uniform)
    return denoised, sigma_lf, freqs, magnitude, lf_mask

```

```

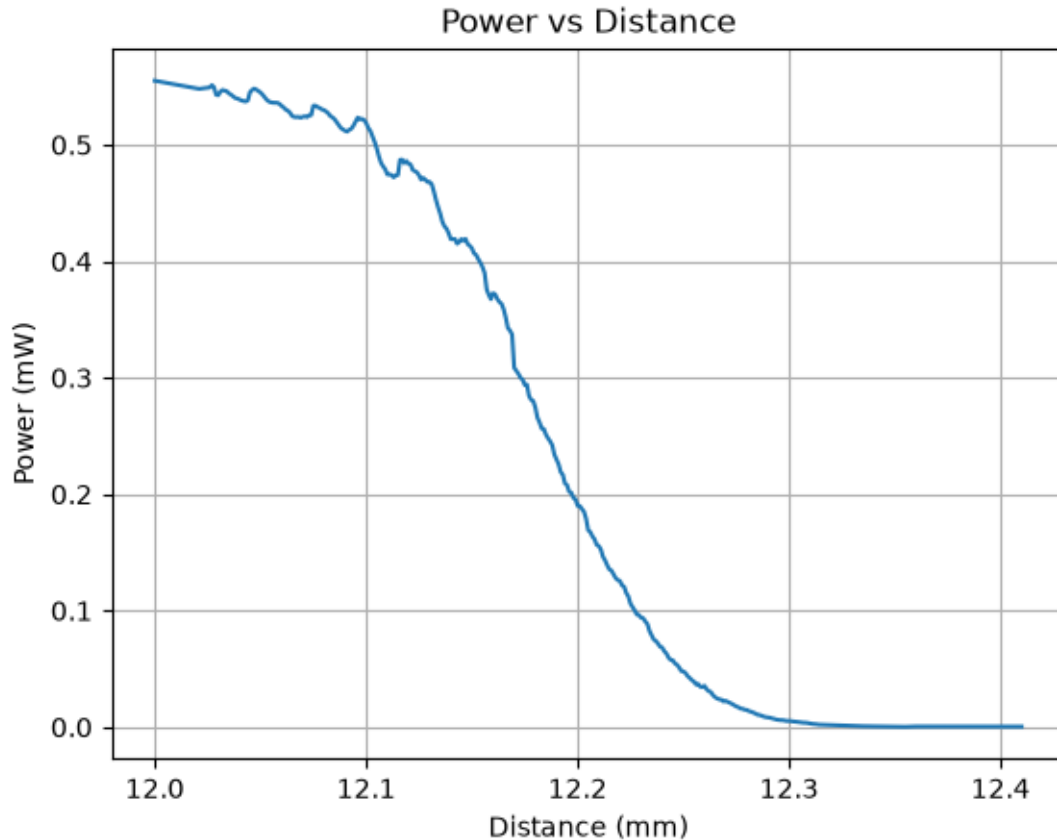
[7]: df = pd.read_csv('data/razor_data.csv')
df['Power (mW)'] = pd.to_numeric(df['Power (mW)'])
df['Distance (mm)'] = pd.to_numeric(df['Distance (mm)'])
df_clean = df.dropna().copy()

```

```
[8]: plt.plot(df_clean['Distance (mm)'], df_clean['Power (mW)'])

plt.title('Power vs Distance')
plt.xlabel('Distance (mm)')
plt.ylabel('Power (mW)')
plt.grid(True)

plt.show()
```



```
[9]: df_clean = df_clean.sort_values('Distance (mm)').reset_index(drop=True)

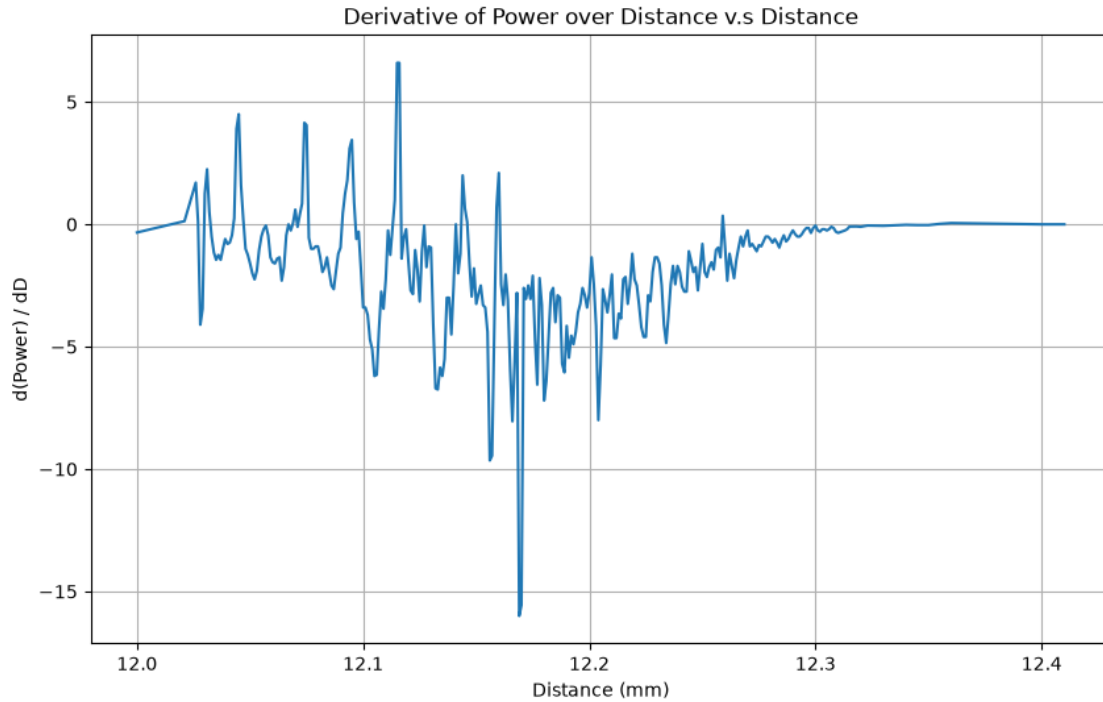
df_clean['Derivative'] = np.gradient(df_clean['Power (mW)'], df_clean['Distance (mm)'])

plt.figure(figsize=(10,6))
plt.plot(df_clean['Distance (mm)'], df_clean['Derivative'])

plt.title('Derivative of Power over Distance v.s Distance')
plt.xlabel('Distance (mm)')
```

```
plt.ylabel('d(Power) / dD')
plt.grid(True)

plt.show()
```



```
[10]: df_clean = df.dropna().sort_values('Distance (mm)').reset_index(drop=True)
# The first k-1 values will result in NaN.
window = 5
rolling_median = df_clean['Power (mW)'].rolling(window=window, center=True).
    ↪median()
df_clean = df_clean.dropna().reset_index(drop=True)

x1 = df_clean['Distance (mm)'].values
y1 = df_clean['Power (mW)'].values
```

```
[16]: def exponential_smooth(signal, alpha=0.25):
    smoothed = pd.Series(signal).ewm(alpha=alpha, adjust=False).mean().values
    return smoothed, alpha
```

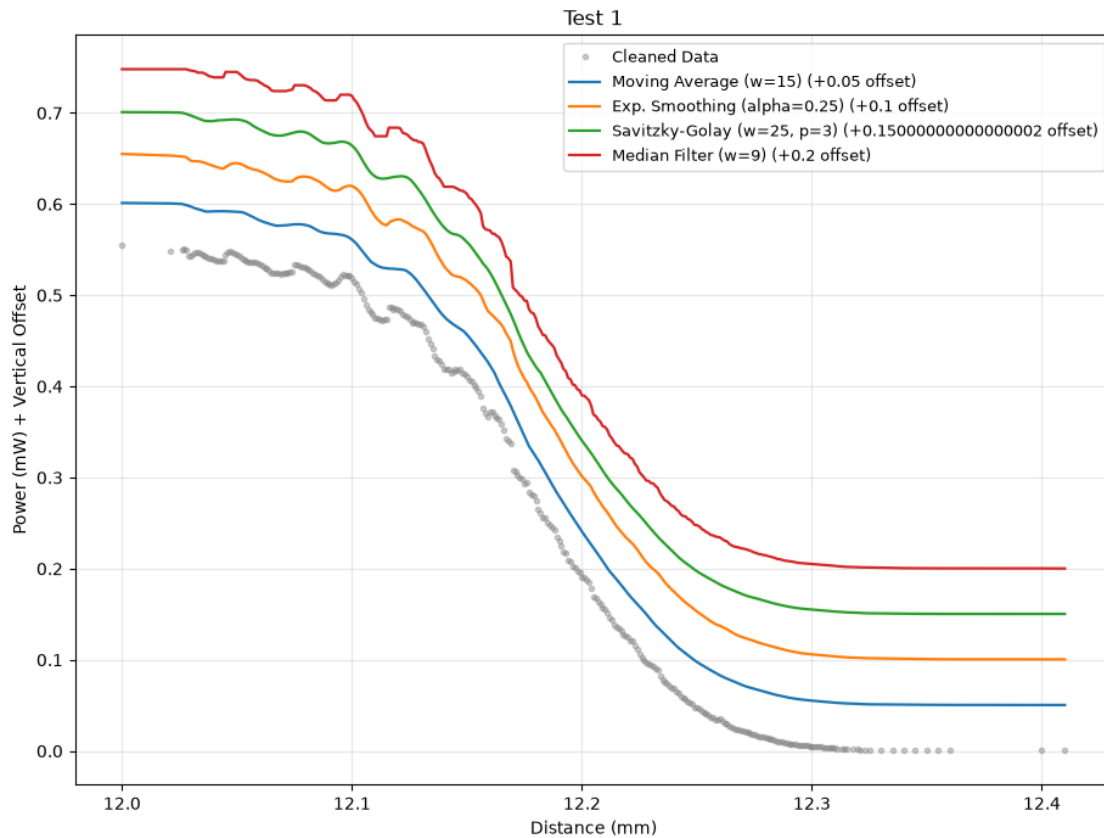
```
[19]: ma1 = moving_average_filter(y1, window=15)
es1, alpha1 = exponential_smooth(y1, alpha=0.25)
sg1 = sg_smooth(y1, window_length=25, polyorder=3)
med1 = median_smooth(y1, window=9)
```

```

scurve_methods_1 = {
    'Moving Average (w=15)': ma1,
    f'Exp. Smoothing (alpha={alpha1:.2f})': es1,
    'Savitzky-Golay (w=25, p=3)': sg1,
    'Median Filter (w=9)': med1,
}

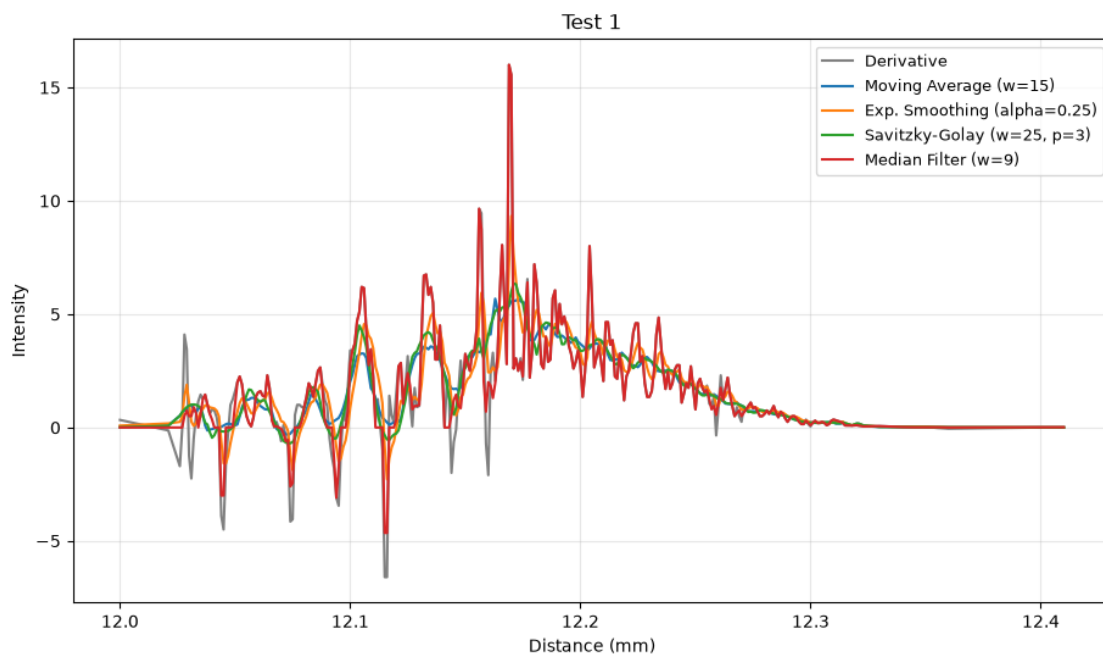
plt.figure(figsize=(11, 8))
plt.plot(x1, y1, '.', color='grey', alpha=0.4, label='Cleaned Data')
offset_step = 0.05
for i, (name, sig) in enumerate(scurve_methods_1.items()):
    current_offset = (i + 1) * offset_step
    plt.plot(x1, sig + current_offset, linewidth=1.6, label=f"{name}_␣
↳(+{current_offset} offset)")
plt.title('Test 1')
plt.xlabel('Distance (mm)')
plt.ylabel('Power (mW) + Vertical Offset')
plt.legend(fontsize=9)
plt.grid(alpha=0.3)
plt.show()

```



```
[20]: raw_profile_1 = -np.gradient(y1, x1)

plt.figure(figsize=(11, 6))
plt.plot(x1, raw_profile_1, color='grey', label='Derivative')
for name, sig in scurve_methods_1.items():
    prof = -np.gradient(sig, x1)
    plt.plot(x1, prof, label=name)
plt.title('Test 1')
plt.xlabel('Distance (mm)')
plt.ylabel('Intensity')
plt.legend(fontsize=9)
plt.grid(alpha=0.3)
plt.show()
```



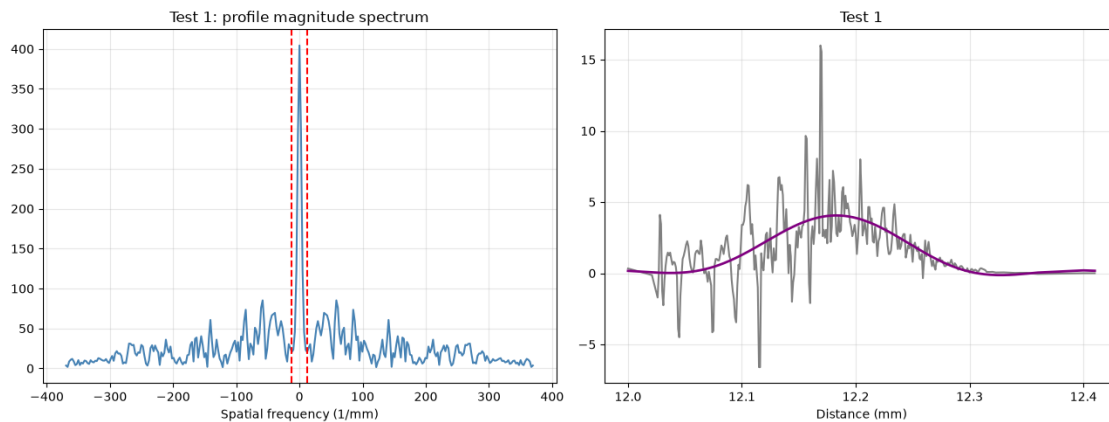
```
[21]: fft_profile_1, sigma_lf_1, freqs_1, mag_1, lf_mask_1 = fft_denoise_profile(x1, raw_profile_1)

fig, axes = plt.subplots(1, 2, figsize=(13, 5))
axes[0].plot(freqs_1, mag_1, color='steelblue')
axes[0].axvline(4 * sigma_lf_1, color='red', linestyle='--')
axes[0].axvline(-4 * sigma_lf_1, color='red', linestyle='--')
axes[0].set_title('Test 1: profile magnitude spectrum')
axes[0].set_xlabel('Spatial frequency (1/mm)')
axes[0].grid(alpha=0.3)
```

```

axes[1].plot(x1, raw_profile_1, color='grey', label='Derivative')
axes[1].plot(x1, fft_profile_1, color='purple', linewidth=2,
    ↳label='FFT-denoised')
axes[1].set_title('Test 1')
axes[1].set_xlabel('Distance (mm)')
axes[1].grid(alpha=0.3)
plt.tight_layout()
plt.show()

```

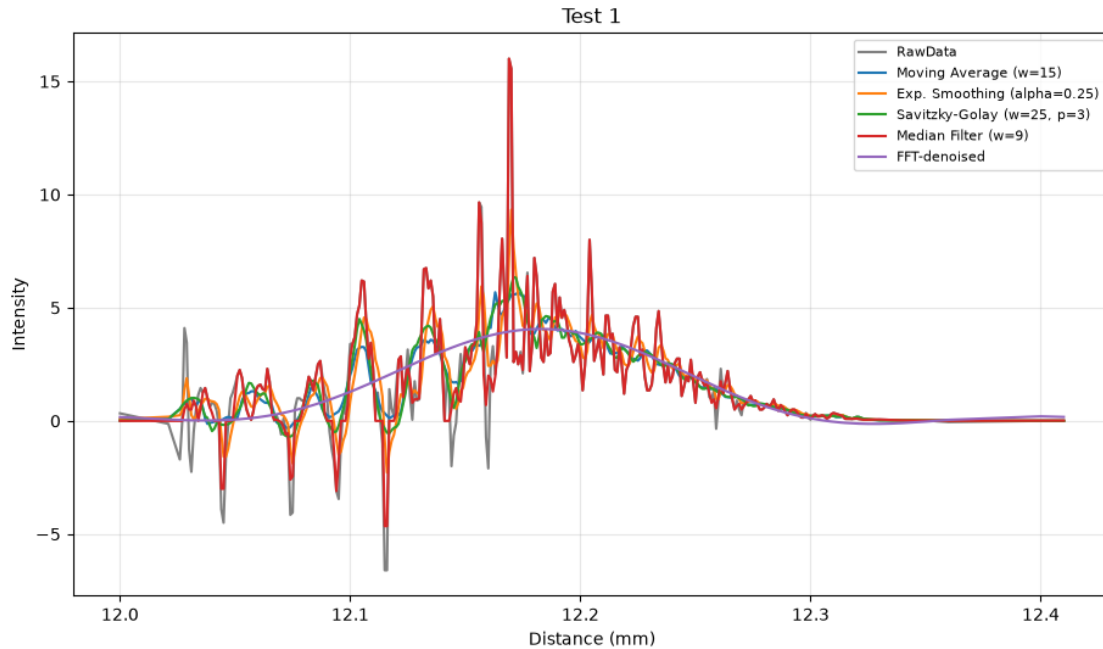


```

[22]: all_profiles_1 = {name: -np.gradient(sig, x1) for name, sig in scurve_methods_1.
    ↳items()}
all_profiles_1['FFT-denoised'] = fft_profile_1

plt.figure(figsize=(11, 6))
plt.plot(x1, raw_profile_1, color='grey', label='RawData')
for name, prof in all_profiles_1.items():
    plt.plot(x1, prof, label=name)
plt.title('Test 1')
plt.xlabel('Distance (mm)')
plt.ylabel('Intensity')
plt.legend(fontsize=8)
plt.grid(alpha=0.3)
plt.show()

```

```
[23]: def gaussian_beam(x, A, x0, w, offset):
        return A * np.exp(-2 * (x - x0) ** 2 / w ** 2) + offset

def fit_beam_radius(x, profile):
    center_guess = np.average(x, weights=profile)
    variance_guess = np.average((x - center_guess)**2, weights=profile)
    std_guess = np.sqrt(variance_guess)
    w_guess = 2 * std_guess
    p0 = [profile.max(), center_guess, w_guess, 0]
    popt, pcov = curve_fit(gaussian_beam, x, profile, p0=p0)
    perr = np.sqrt(np.diag(pcov))
    return popt, perr
```

```
[24]: profiles_1 = {
    'Moving Average': -np.gradient(ma1, x1),
    'Exp. Smoothing': -np.gradient(es1, x1),
    'Savitzky-Golay': -np.gradient(sg1, x1),
    'Median Filter': -np.gradient(med1, x1),
    'FFT-denoised': fft_profile_1,
}

print(f"{'Method':16s}{'w (1/e^2 radius, um)':>24s}{'center (mm)':>16s}")
fits_1 = {}
for name, prof in profiles_1.items():
    popt, perr = fit_beam_radius(x1, prof)
```

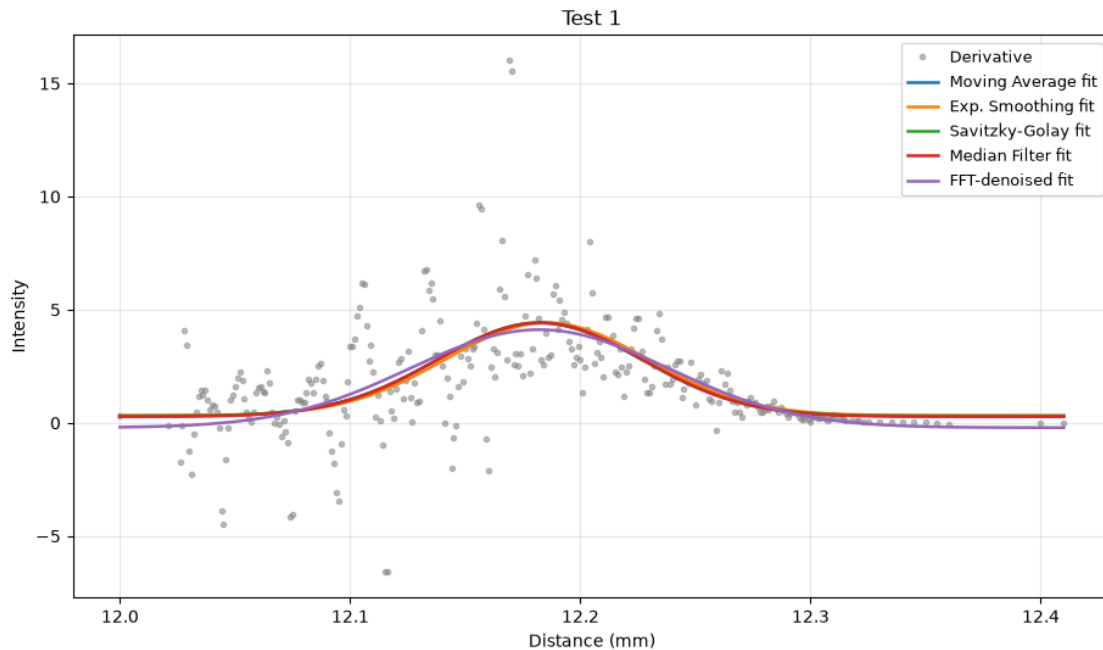
```

A, x0, w, off = popt
fits_1[name] = popt
print(f"{name:16s}{abs(w)*1000:16.2f} +/- {perr[2]*1000:5.2f}{x0:16.4f}")

plt.figure(figsize=(11, 6))
plt.plot(x1, raw_profile_1, '.', color='grey', alpha=0.5, label='Derivative')
x1_fine = np.linspace(x1.min(), x1.max(), 500)
for name, popt in fits_1.items():
    plt.plot(x1_fine, gaussian_beam(x1_fine, *popt), linewidth=1.8,
             label=f'{name} fit')
plt.title('Test 1')
plt.xlabel('Distance (mm)')
plt.ylabel('Intensity')
plt.legend(fontsize=9)
plt.grid(alpha=0.3)
plt.show()

```

Method	w (1/e ² radius, μm)	center (mm)
Moving Average	88.79 +/- 3.13	12.1824
Exp. Smoothing	87.91 +/- 5.16	12.1854
Savitzky-Golay	88.25 +/- 4.20	12.1824
Median Filter	89.18 +/- 8.84	12.1824
FFT-denoised	111.99 +/- 0.54	12.1821



2 6/29 Razor Test 2

2.0.1 Notes:

Collimator face to reflection off wedge 85mm

Wedge to mirror 133mm

Mirror to lens 102.5mm

Lens to blade 56 mm

Blade to sensor 200mm

Focal length 50mm 1in dia

Computational dist: | d1: 326.57 (320.5 face to d1) | d2: 55.32 | Path: 375.82

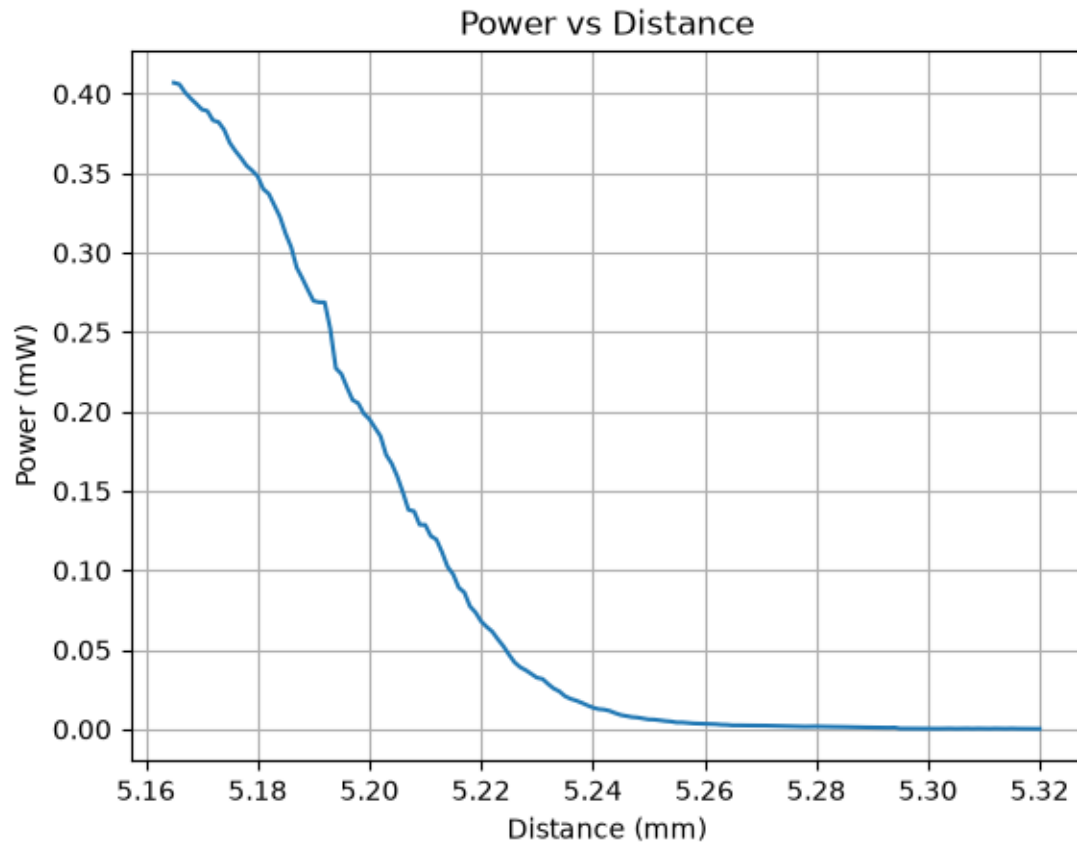
Measured distances: | d1: (320.5) | d2: 56 | Path: 376.5

```
[25]: df2 = pd.read_csv('data/razor_data2.csv')
df2['Power (mW)'] = pd.to_numeric(df2['Power (mW)'])
df2['Distance (mm)'] = pd.to_numeric(df2['Distance (mm)'])
df2_clean = df2.dropna().copy()
```

```
[26]: plt.plot(df2_clean['Distance (mm)'], df2_clean['Power (mW)'])

plt.title('Power vs Distance')
plt.xlabel('Distance (mm)')
plt.ylabel('Power (mW)')
plt.grid(True)

plt.show()
```



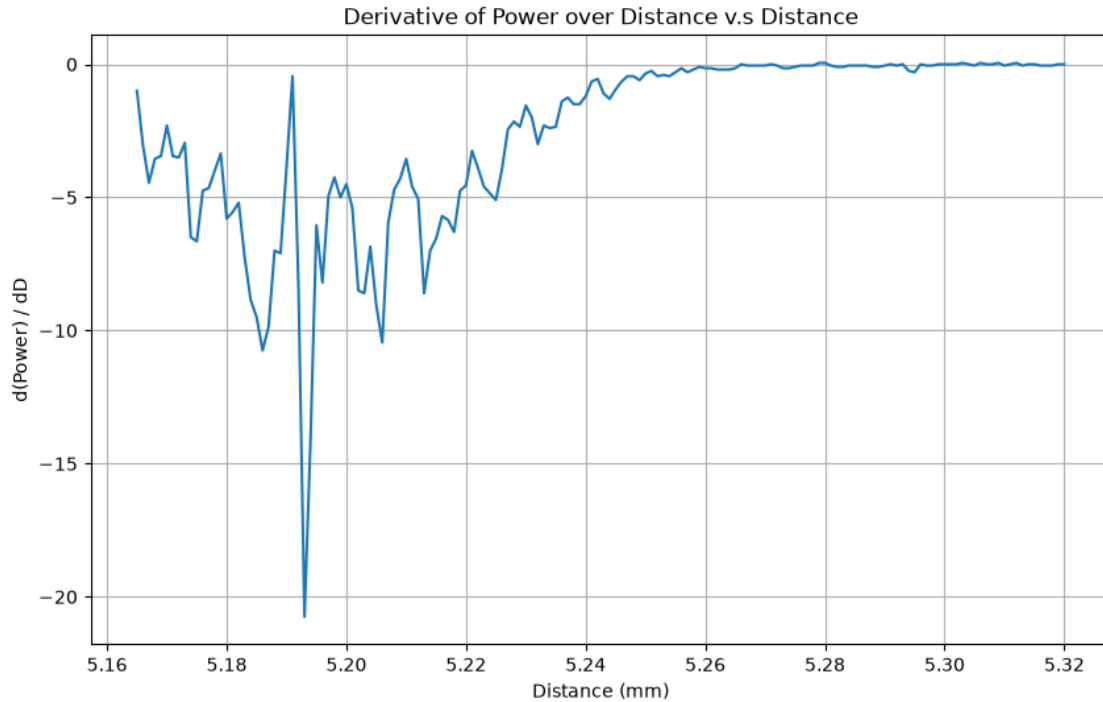
```
[27]: df2_clean = df2_clean.sort_values('Distance (mm)').reset_index(drop=True)

df2_clean['Derivative'] = np.gradient(df2_clean['Power (mW)'],
    ↪df2_clean['Distance (mm)'])

plt.figure(figsize=(10,6))
plt.plot(df2_clean['Distance (mm)'], df2_clean['Derivative'])

plt.title('Derivative of Power over Distance v.s Distance')
plt.xlabel('Distance (mm)')
plt.ylabel('d(Power) / dD')
plt.grid(True)

plt.show()
```



```
[28]: df2_clean = df2.dropna().sort_values('Distance (mm)').reset_index(drop=True)

window = 5
rolling_median = df2_clean['Power (mW)'].rolling(window=window, center=True).
    ↪median()
df2_clean = df2_clean.dropna().reset_index(drop=True)

x2 = df2_clean['Distance (mm)'].values
y2 = df2_clean['Power (mW)'].values
```

```
[30]: ma2 = moving_average_filter(y2, window=9)
es2, alpha2 = exponential_smooth(y2, alpha=0.25)
sg2 = sg_smooth(y2, window_length=15, polyorder=3)
med2 = median_smooth(y2, window=7)

scurve_methods_2 = {
    'Moving Average (w=9)': ma2,
    f'Exp. Smoothing (alpha={alpha2:.2f})': es2,
    'Savitzky-Golay (w=15, p=3)': sg2,
    'Median Filter (w=7)': med2,
}

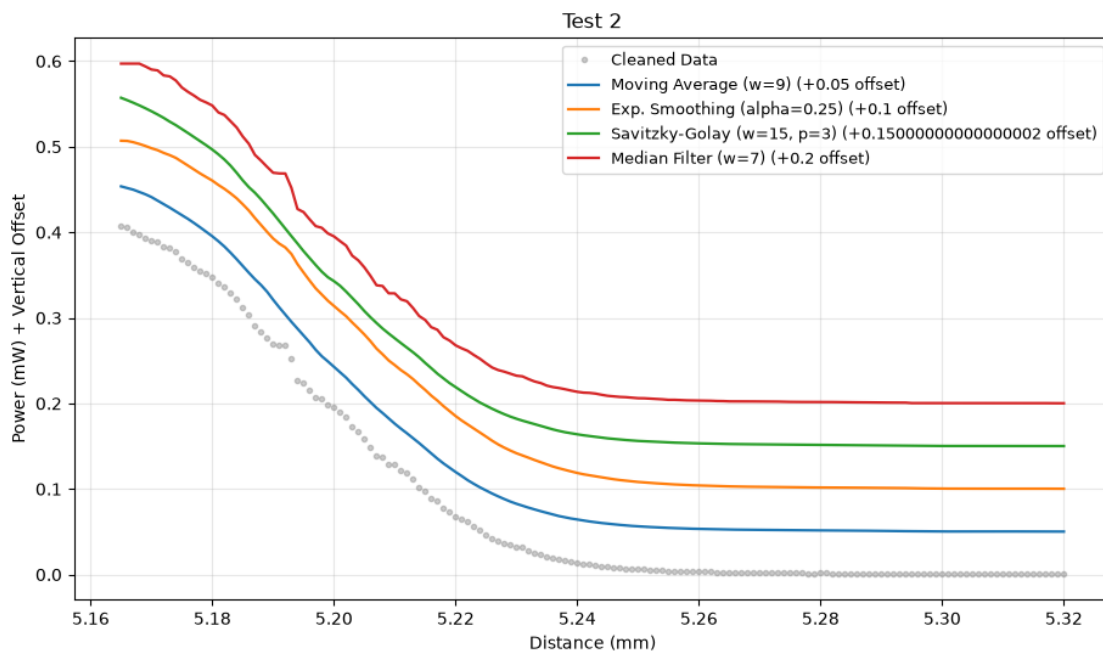
plt.figure(figsize=(11, 6))
plt.plot(x2, y2, '.', color='grey', alpha=0.4, label='Cleaned Data')
```

```

offset_step = 0.05
for i, (name, sig) in enumerate(scurve_methods_2.items()):
    current_offset = (i + 1) * offset_step
    plt.plot(x2, sig + current_offset, linewidth=1.6, label=f"{name}_␣
    ↳(+{current_offset} offset)")

plt.title('Test 2')
plt.xlabel('Distance (mm)')
plt.ylabel('Power (mW) + Vertical Offset')
plt.legend(fontsize=9)
plt.grid(alpha=0.3)
plt.show()

```



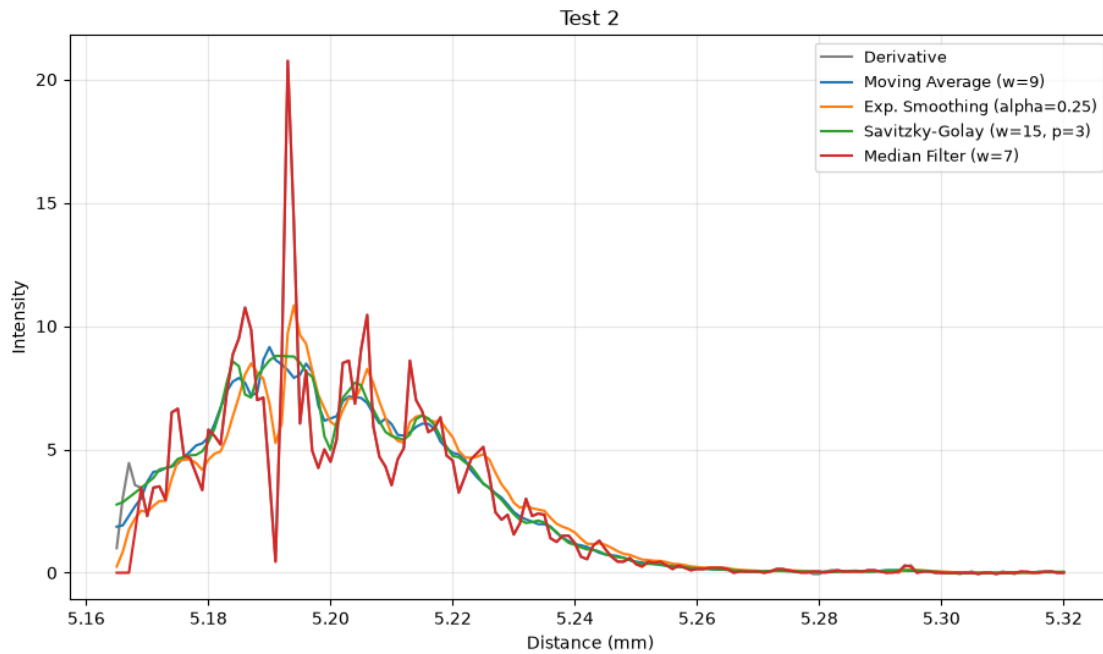
```

[32]: raw_profile_2 = -np.gradient(y2, x2)

plt.figure(figsize=(11, 6))
plt.plot(x2, raw_profile_2, color='grey', label='Derivative')
for name, sig in scurve_methods_2.items():
    prof = -np.gradient(sig, x2)
    plt.plot(x2, prof, label=name)
plt.title('Test 2')
plt.xlabel('Distance (mm)')
plt.ylabel('Intensity')
plt.legend(fontsize=9)
plt.grid(alpha=0.3)

```

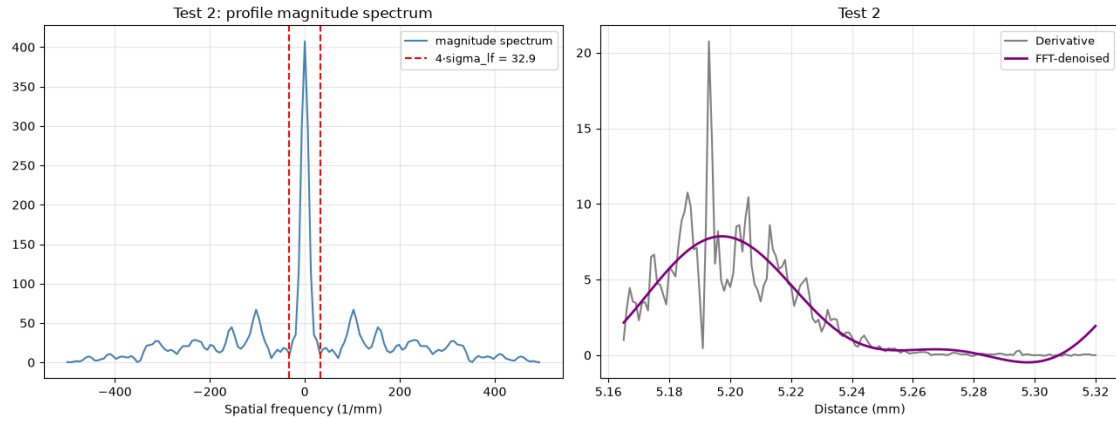
```
plt.show()
```



```
[33]: fft_profile_2, sigma_lf_2, freqs_2, mag_2, lf_mask_2 = fft_denoise_profile(x2, raw_profile_2)

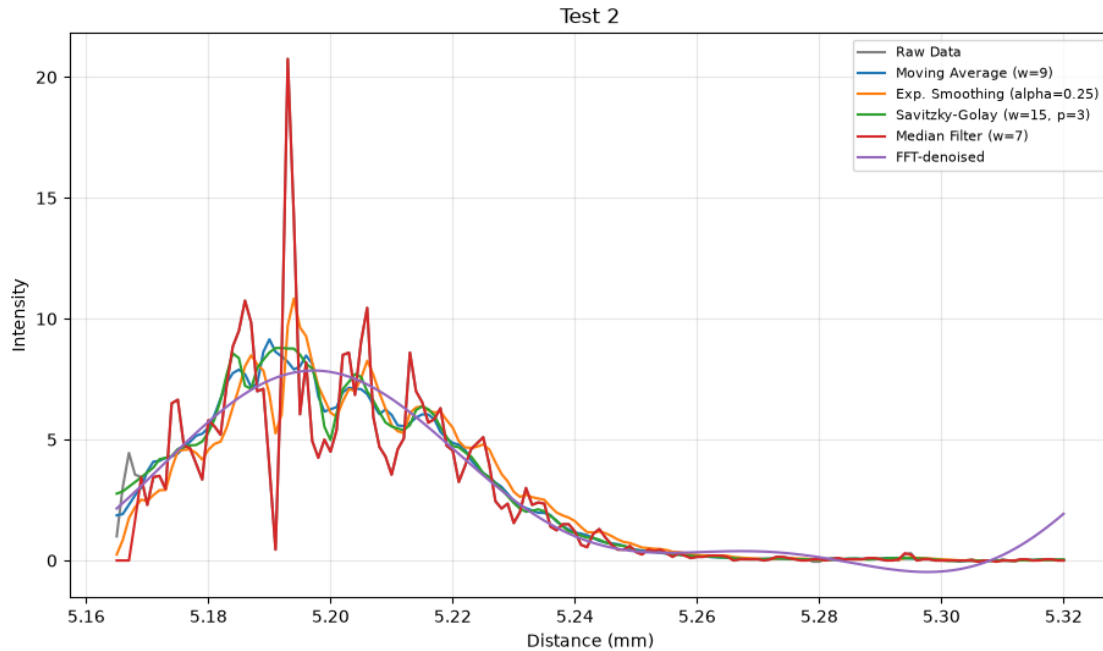
fig, axes = plt.subplots(1, 2, figsize=(13, 5))
axes[0].plot(freqs_2, mag_2, color='steelblue', label='magnitude spectrum')
axes[0].axvline(4 * sigma_lf_2, color='red', linestyle='--', label=f'4 * sigma_lf_2')
axes[0].axvline(-4 * sigma_lf_2, color='red', linestyle='--')
axes[0].set_title('Test 2: profile magnitude spectrum')
axes[0].set_xlabel('Spatial frequency (1/mm)')
axes[0].legend(fontsize=9)
axes[0].grid(alpha=0.3)

axes[1].plot(x2, raw_profile_2, color='grey', label='Derivative')
axes[1].plot(x2, fft_profile_2, color='purple', linewidth=2, label='FFT-denoised')
axes[1].set_title('Test 2')
axes[1].set_xlabel('Distance (mm)')
axes[1].legend(fontsize=9)
axes[1].grid(alpha=0.3)
plt.tight_layout()
plt.show()
```



```
[34]: all_profiles_2 = {name: -np.gradient(sig, x2) for name, sig in scurve_methods_2.
      ↪ items()}
all_profiles_2['FFT-denoised'] = fft_profile_2

plt.figure(figsize=(11, 6))
plt.plot(x2, raw_profile_2, color='grey', label='Raw Data')
for name, prof in all_profiles_2.items():
    plt.plot(x2, prof, label=name)
plt.title('Test 2')
plt.xlabel('Distance (mm)')
plt.ylabel('Intensity')
plt.legend(fontsize=8)
plt.grid(alpha=0.3)
plt.show()
```

```
[35]: profiles_2 = {
    'Moving Average': -np.gradient(ma2, x2),
    'Exp. Smoothing': -np.gradient(es2, x2),
    'Savitzky-Golay': -np.gradient(sg2, x2),
    'Median Filter': -np.gradient(med2, x2),
    'FFT-denoised': fft_profile_2,
}

print(f"{'Method':16s}{'w (1/e^2 radius, um)':>24s}{'center (mm)':>16s}")
fits_2 = {}
for name, prof in profiles_2.items():
    popt, perr = fit_beam_radius(x2, prof)
    A, x0, w, off = popt
    fits_2[name] = popt
    print(f"{'name':16s}{'abs(w)*1000:16.2f'} +/- {'perr[2]*1000:5.2f'}{'x0:16.4f}")

plt.figure(figsize=(11, 6))
plt.plot(x2, raw_profile_2, '.', color='grey', alpha=0.5, label='Derivative')
x2_fine = np.linspace(x2.min(), x2.max(), 500)
for name, popt in fits_2.items():
    plt.plot(x2_fine, gaussian_beam(x2_fine, *popt), linewidth=1.8,
             label=f'{name} fit')
plt.title('Test 2')
plt.xlabel('Distance (mm)')
plt.ylabel('Intensity')
```

```
plt.legend(fontsize=9)
plt.grid(alpha=0.3)
plt.show()
```

Method	w (1/e ² radius, μm)	center (mm)
Moving Average	43.91 +/- 0.76	5.1968
Exp. Smoothing	43.19 +/- 1.13	5.2001
Savitzky-Golay	44.56 +/- 0.94	5.1964
Median Filter	41.69 +/- 2.83	5.1975
FFT-denoised	41.69 +/- 0.57	5.1976

